

Automated Problem-to-Solution Generation for a Tokamak Fusion-Plasma Simulation

A hierarchical multi-agent PETSc system: from a plain-language prompt to a verified Grad-Shafranov solver
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1 Motivation

Correct, fast HPC simulation code still demands scarce, implicit expertise in numerical methods and library APIs.
Can a multi-agent AI system automate the path from a plain-language idea to VERIFIED code for a real fusion problem?
We apply the open PETSc multi-agent system to the tokamak Grad-Shafranov equilibrium.

2 The problem

Axisymmetric ideal-MHD force balance for the poloidal flux $\psi(R,Z)$:

$\Delta \psi = -\mu_0 R^2 p'(\psi) - F F'(\psi)$

Gives the flux surfaces, magnetic axis, and safety factor q .
Elliptic, nonlinear → PETSc SNES; verifiable vs an exact solution.

3 The multi-agent system

Three layers of specialist agents (MCP servers):

- Modeling → PDE identity & strong/weak forms
- Numerical Analysis → grid, discretization, solver
- HPC Code Generation → writes, compiles & runs PETSc C
- Driver → records every artifact (provenance)

All agents run on ANL's Argo gateway (Claude Opus 4.8).

4 The pipeline in action

Prompt: “the Grad-Shafranov equilibrium for the plasma in a tokamak.”
Modeling → ‘Grad-Shafranov equation’, steady.
Numerical Analysis → nonlinear → SNES.
Code Generation → 267-line PETSc DMDA+SNES solver.
Compiled & ran on 1 and 4 MPI ranks — no human edits.

5 Generated solver (excerpt)

```
/* residual: Delta*_h psi - f */
dRR=(x[j][i+1]-2x[j][i]+x[j][i-1])/hR2;
dZZ=(x[j+1][i]-2x[j][i]+x[j-1][i])/hZ2;
f[j][i]=dRR-dR/R+dZZ-ForcingF(u,R,Z);
SNESSetJacobian(snes,J,J,FormJacobian,&u);
SNESolve(snes,NULL,x);
```

6 Verification

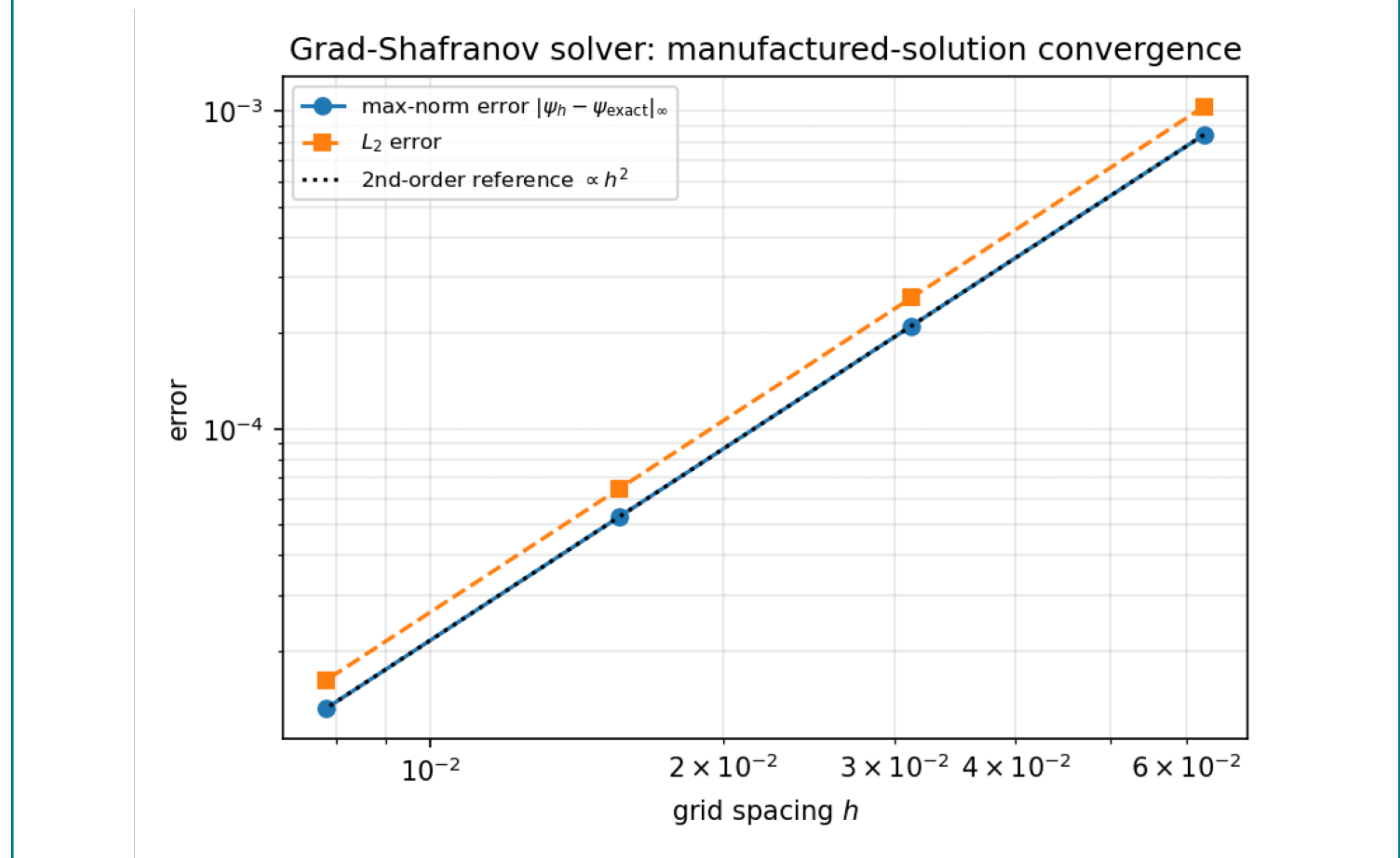


Fig. 2 Error vs grid spacing tracks $h^2 \rightarrow$ observed order $p = 2.00, 2.00, 2.00$.

7 Decision-gate metrics

Model identified ✓ Compiled & ran ✓ SNES converged ✓
Second-order: $p = 2.00, 2.00, 2.00$ (error $8.4e-04 \rightarrow 1.3e-05$, $33 \rightarrow 257$).
Human effort: 0 solver lines written; 267 generated.
Efficiency: ~450.8 s wall-clock; ~23 LLM completions.

8 Contributions & conclusions

A verified PETSc fusion-MHD solver produced from a prompt.
Verification-driven: exact-solution + convergence checks.
Upstream fixes: CWD-independent servers; graceful no-docs/RAG; reliable capture.
github.com/engineer-scientist/petsc_mcp_servers_tokamak

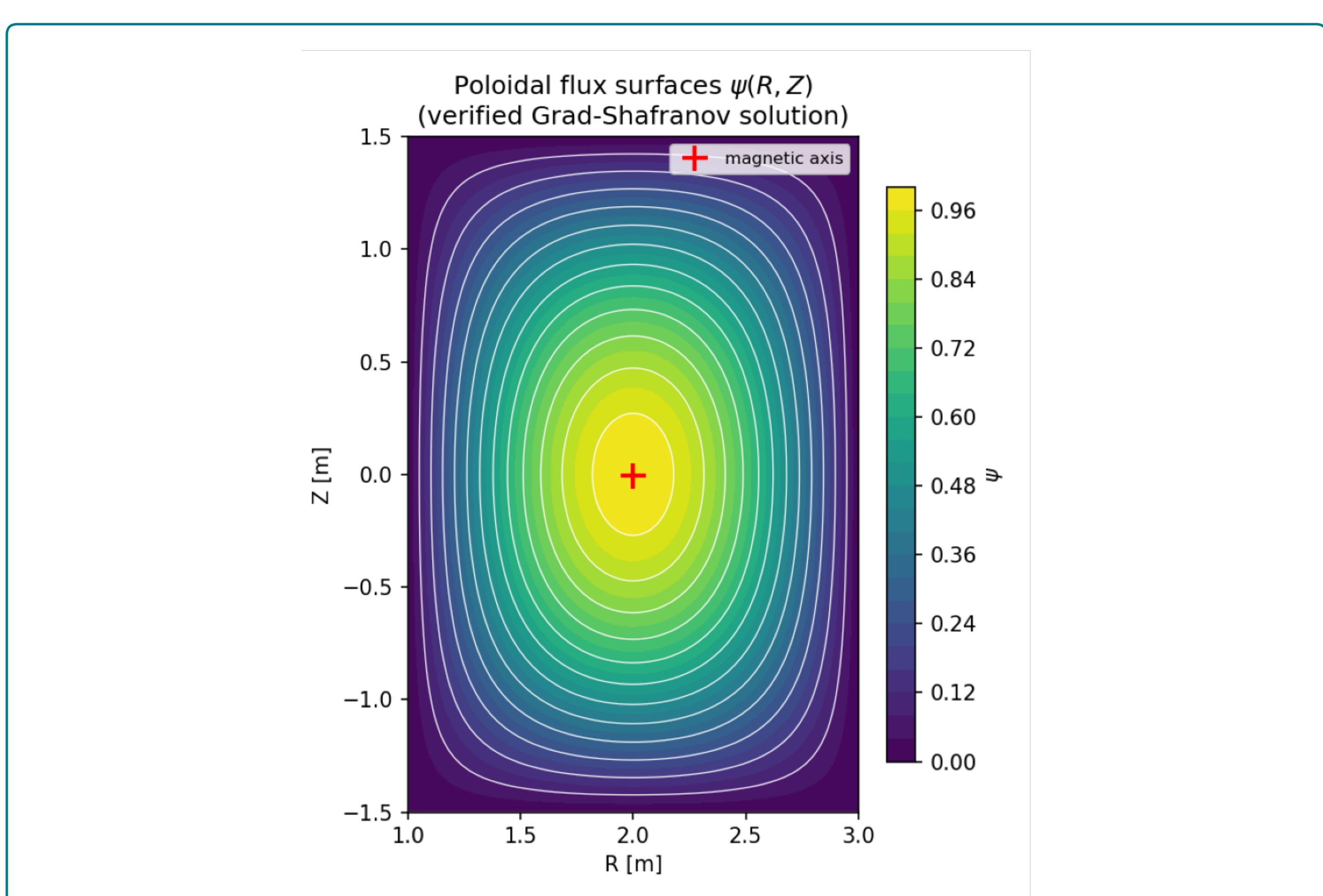


Fig. 1 Poloidal flux surfaces $\psi(R,Z)$ of the verified equilibrium.